Day 1 Reference Guide: Cloud Basics, Careers, Certification and Account Security

Cloud Unlock · Saturday, 10 October 2026 · Services, key ideas and official AWS links

---

Live class: Saturday 10 Oct 2026, 7:00-9:00 PM IST
Exam focus: Domain 1 (Cloud Concepts), Domain 2 (Security), Domain 4 (Billing)
Exam tasks: 1.1 · 2.3 · 4.2 (plus 1.2 and 1.4 as reading)

Today covers: cloud careers and jobs · AWS certifications · opportunities · networking · learning and projects · live Q&A · and your first practical: a safe AWS account.

What You Will Know

- You can explain cloud computing using one Amazon festival sale night: scalability, elasticity, high availability, reliability, security and pay-per-use.
- You know four cloud roles and the skills fresher jobs ask for.
- You know the AWS certification levels and the CLF-C02 exam format.
- You know two free places to learn AWS and two ways to meet the cloud community.
- Read after class: cloud economics, IaaS/PaaS/SaaS and the six pillars (taught again in S7, S10 and S11).
- You can sign in, switch Regions and find your Free Tier or credits page.
- Your root user has MFA, and you have an IAM admin user for daily work.
- Also shown today: Region, Availability Zone, edge location, Local Zone and Wavelength Zone (terms).
- You can name tasks only the root user can do, and tell Budgets, Cost Explorer and Pricing Calculator apart.

Day 1 - Cloud Foundations

Goal: understand what the cloud is and which real problems it solves.

Exam tasks: 1.1 (benefits of the AWS Cloud), 1.4 (cloud economics), 3.1 (deployment models)

The Story - Sale Night On A Shopping App

Think of **Flipkart Big Billion Days**, the **Amazon Great Indian Festival** or a **Myntra sale**. When the sale opens, people all over India open the app at the same moment. They search, add to cart and pay with **PhonePe** or **Google Pay**. A week later the rush is gone.

Now imagine **you** run an app like that. These are teaching numbers, not real company figures:

When	Shoppers on the app	Servers you need
Normal day	A few thousand	2
Sale opens	Lakhs at once	20
A week later	A few thousand again	2

With your own servers: you buy 20 servers for the sale, wait weeks for them to arrive, and pay for all 20 all year. For 50 weeks, 18 of them sit idle.

With the cloud: you add servers when the sale opens, remove them when it ends, and pay only for the hours you used them.

Walk through one night - Amazon Great Indian Festival

You have been waiting for a phone deal. Follow your own steps and see what the app has to do at each one:

Your step	What the app must do	Cloud word
11:59 PM - you and lakhs of others open the app and keep refreshing	Handle a sudden rush without slowing down	Scalability, elasticity
12:00 AM - the deal opens, you tap **Add to Cart**	Keep your cart even if one server fails	High availability
You pay with **PhonePe** or **Google Pay**	Never lose the order or the payment record	Reliability
You open **My Orders**	Show your orders and address only to you	Security
A week later the sale ends	Remove the extra servers and stop paying for them	Cost (pay for what you use)

The same thing happens in **Flipkart Big Billion Days** and the **Myntra** sale.

These five steps are the five problems the cloud solves (next section). We describe only what you see as a user, not how these companies build their systems.

Five Problems The Cloud Solves

Problem	Without cloud	With AWS	Service (later)
Scaling (autoscaling)	Buy for the peak	Capacity grows and shrinks with traffic	EC2 Auto Scaling, Elastic Load Balancing
Cost	Large upfront spend	Pay for what you use	Budgets, Cost Explorer
Availability	One data centre fails, site down	Spread across Availability Zones	Multi-AZ, Route 53, CloudFront
Security	You secure everything alone	Shared responsibility	IAM, KMS, Security Groups
Reliability	Manual backups	Automated backups and replication	S3, RDS, AWS Backup

Six Advantages Of Cloud Computing

1. Trade fixed expense for variable expense.
2. Benefit from massive economies of scale.
3. Stop guessing capacity.
4. Increase speed and agility.
5. Stop spending money running and maintaining data centres.
6. Go global in minutes.

Four Words The Exam Loves

Word	Meaning	Real-app example
High availability	The app keeps running when one part fails	The Amazon app still opens on sale night if one data centre has a problem
Elasticity	Resources grow **and shrink** automatically with demand	2 servers normally, 20 when the sale opens, back to 2 after
Scalability	The system can handle more load by adding resources	Ready for more shoppers as the app grows every year
Agility	Try new things fast and cheaply	Try a new "Buy Now" button for one week, remove it if nobody uses it

Global reach is a benefit too: deploy close to users anywhere in minutes.

Cloud Economics

Idea	On-premises	AWS Cloud
Fixed vs variable cost	Big upfront spend (CapEx) on servers, buildings, power, cooling, staff	Pay for what you use (OpEx)

Idea	On-premises	AWS Cloud
Hidden on-prem costs	Hardware refresh, electricity, cooling, floor space, security guards, data centre staff	Included in the service price
Economies of scale	Your small volume, your price	AWS buys at huge volume and passes savings on
Rightsizing	Hard: you already bought the server	Pick the right size, change it any time
Automation	Manual setup, slow, error-prone	Scripts and IaC: faster, repeatable, fewer mistakes
Licensing	You own licences	**BYOL** (bring your own licence) or **licence included** in the price

Exam pointer:

- "Reduce upfront cost" or "trade capital expense for variable expense" = cloud economics.
- "Resources match demand automatically" = elasticity.
- "Already own Windows or Oracle licences" = BYOL.
- "Instance is too big for the workload" = rightsizing.

Service Models

- **IaaS**: you manage OS and apps (Amazon EC2).
- **PaaS**: you manage app and data; the platform is managed (Elastic Beanstalk, RDS).
- **SaaS**: you just use it (Gmail, Microsoft 365).

Deployment Models

- Cloud, on-premises (private cloud), and hybrid (both together).

Exam pointer:

- Pay-as-you-go and no upfront cost point to cloud benefits.
- Own data centre plus AWS together means hybrid.
- You manage the OS means IaaS.

Official Links

Topic	Link
What is cloud computing	https://aws.amazon.com/what-is-cloud-computing/
Types of cloud computing (IaaS, PaaS, SaaS)	https://aws.amazon.com/types-of-cloud-computing/
Six advantages of cloud computing	https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html
AWS customer stories	https://aws.amazon.com/solutions/case-studies/
AWS Pricing overview	https://aws.amazon.com/pricing/

Day 1 - Careers, Certifications, Opportunities and Networking

Goal: know where cloud skills can take you, which certification to start with, where to learn for free and how to meet people in the cloud community.

Certification Path

- Foundational: **Cloud Practitioner (CLF-C02)** (our goal), AI Practitioner.
- Associate: Solutions Architect, Developer, SysOps/CloudOps Engineer, Data Engineer.
- Professional and Specialty: senior roles and specialisation.
  Check aws.amazon.com/certification for the current list, prices and renewal rules.

The CLF-C02 Exam At A Glance

Item	Detail
Questions	65 (50 scored + 15 unscored; you cannot tell which)
Time	90 minutes
Question types	Multiple choice (one answer) and multiple response (two or more answers)
Pass mark	700 out of 1000 (scaled score); you only need to pass overall
Guessing	No penalty, so never leave a question blank
Domains	Cloud Concepts 24% · Security and Compliance 30% · Cloud Technology and Services 34% · Billing, Pricing and Support 12%

Cloud Careers And Jobs

Role	What you do	Common starting skills
Cloud Support Associate	Help customers fix cloud problems	AWS basics, Linux, networking
Cloud Engineer	Build and run systems on AWS	AWS, Linux, scripting
DevOps Engineer	Automate how apps are built and released	Git, CI/CD, containers, IaC
Solutions Architect	Design the whole system	Broad AWS knowledge, communication

Data Engineer and Security Engineer are other common paths.

Employers look for hands-on skills plus a certification. That is why every week has a practical and a GitHub submission: it becomes your portfolio.

Opportunities For Students

Opportunity	What it is	Link
AWS Skill Builder	Free AWS courses and practice questions	https://skillbuilder.aws/
AWS Educate	Free cloud learning for students	https://aws.amazon.com/education/awseducate/
AWS Builder Center	Read and share articles by builders	https://builder.aws.com/
AWS Community Builders	A programme for people who build and share what they learn	https://aws.amazon.com/developer/community/community-builders/
Your own projects	A website for your college club or event, built on AWS	Your GitHub

Read each programme's current rules on its official page.

Networking

- **LinkedIn:** post what you learn every week, in your own words, with `#CloudUnlock`. Connect with classmates.
- **AWS communities:** AWS User Groups, AWS Community Day, AWS Summits. Go to one event this year and talk to three people.
- **This group:** the Cloud Unlock WhatsApp group, Meetup and YouTube live chat.
- **Ask good questions:** AWS re:Post, the official AWS Q&A community.

Read After Class - The Six Pillars Of The AWS Well-Architected Framework

We teach this in detail in S10. Read it now so the names are familiar (exam task 1.2).

Pillar	Question it asks	Benefit	Example
Operational Excellence	Can we run, monitor and improve it?	Fewer outages	Automate deployments, monitor
Security	Is data and access protected?	Trust, compliance	Least-privilege IAM, encryption
Reliability	Does it recover and keep working?	Less downtime	Multi-AZ, backups
Performance Efficiency	Do we use the right resources?	Fast, no waste	Right instance type, caching
Cost Optimization	Do we pay only for value?	Lower bills	Right-size, turn off idle
Sustainability	Do we minimise environmental impact?	Lower energy and cost	Efficient resources, scale down

Exam pointer: "recover from failure" = Reliability. "Reduce spend" = Cost Optimization. "Protect data" = Security.

Common confusion:

- Performance Efficiency is about using the **right** resources well (speed). Cost Optimization is about **not paying** for waste.
- Operational Excellence is about **running and improving** the workload. Reliability is about **recovering** from failure.
- Sustainability is about **energy and environmental impact**, often by removing idle resources.

Official Links

Topic	Link
Six pillars of the Well-Architected Framework	https://docs.aws.amazon.com/wellarchitected/latest/framework/the-pillars-of-the-framework.html
AWS Well-Architected overview	https://aws.amazon.com/architecture/well-architected/
AWS Educate	https://aws.amazon.com/education/awseducate/
AWS Community Builders	https://aws.amazon.com/developer/community/community-builders/
AWS User Groups	https://aws.amazon.com/developer/community/usergroups/
AWS Community Day	https://aws.amazon.com/events/community-day/
AWS re:Post	https://repost.aws/
AWS Certification (all levels)	https://aws.amazon.com/certification/
AWS Certified Cloud Practitioner (exam page and guide)	https://aws.amazon.com/certification/certified-cloud-practitioner/
AWS Skill Builder (free training)	https://skillbuilder.aws/
AWS Training and Certification	https://aws.amazon.com/training/

More Official Links

Topic	Link
Create and activate an AWS account	https://repost.aws/knowledge-center/create-and-activate-aws-account
Getting started with your AWS account	https://docs.aws.amazon.com/accounts/latest/reference/getting-started.html
AWS Free Tier	https://aws.amazon.com/free/
AWS Free Tier plans and credits	https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/free-tier-plans.html
AWS Management Console guide	https://docs.aws.amazon.com/awsconsolehelpdocs/latest/gsg/what-is.html
AWS Support plans	https://aws.amazon.com/premiumsupport/plans/

Day 1 - Account Security Lab

Goal: create a safe AWS account foundation for the next five weeks.

Exam tasks: 2.3 (root user protection, MFA, IAM users and groups), 4.2 (AWS Budgets)

Why The Root User Is Special

The root user is the email you signed up with. It has full access to everything and cannot be limited by IAM policies.

Some tasks only the root user can do, for example:

- Change account settings: account name, root email, root password
- Close the AWS account
- Change or cancel the AWS Support plan
- Restore permissions when the only IAM admin is locked out
- Turn on IAM user access to the Billing console

Everything else should be done with an IAM user (Lab 3 below) or an IAM role (Week 2).

Exam pointer: "protect the root user" = turn on MFA, do not create root access keys, do not use root for daily work, use a strong unique password.

Budgets Or Cost Explorer?

Tool	**Use it to**	**Think of it as**
AWS Budgets	Get an alert **before or when** spend crosses a limit	Smoke alarm
AWS Cost Explorer	See and analyse **past** spend and forecasts	Bank statement
AWS Pricing Calculator	Estimate cost **before** you build	Shopping quote

Official Links

Topic	**Link**
Root user best practices	https://docs.aws.amazon.com/IAM/latest/UserGuide/root-user-best-practices.html
Tasks that need the root user	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_root-user.html
Turn on MFA for the root user	https://docs.aws.amazon.com/IAM/latest/UserGuide/enable-virt-mfa-for-root.html
MFA in AWS	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa.html
Create an IAM user	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_users_create.html
Create IAM groups	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_groups_create.html

Topic	Link
Add or remove users in a group	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_groups_manage_add-remove-users.html
MFA for an IAM user	https://docs.aws.amazon.com/IAM/latest/UserGuide/enable-virt-mfa-for-iam-user.html
Account alias	https://docs.aws.amazon.com/IAM/latest/UserGuide/console-account-alias.html
IAM best practices	https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html
AWS Budgets	https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html
AWS Cost Explorer	https://docs.aws.amazon.com/cost-management/latest/userguide/ce-what-is.html
AWS Pricing Calculator	https://calculator.aws/
